QML Lab • research project 05

QAOA for MaxCut with Learned Initial Angles

Predicting good QAOA parameters from graph features to cut optimisation cost.

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Quadrant	QC (quantum data or quantum algorithm, classical learner)	**Track**	Variational
Difficulty	Medium	**Tools**	PennyLane, NetworkX, scikit-learn
Builds on	Modules 3 and 5 of the QML Lab	**Workspace**	`qml_projects/qaoa-maxcut-learned-init/`

Summary. The Quantum Approximate Optimization Algorithm (QAOA) solves combinatorial problems such as MaxCut with a short parameterised circuit, but finding good angles costs many circuit evaluations. Optimal angles are known to concentrate across similar graphs. This project trains a classical model that maps simple graph features to good starting angles, and measures how many optimisation steps that saves. The learner is classical and learns about a quantum algorithm, so the project sits on the QC side of the taxonomy.

1 Background and key concepts

1.1 MaxCut

For a graph $G = (V, E)$, MaxCut asks for a split of the vertices into two sets that cuts as many edges as possible. With spins $z_i \in \{-1, +1\}$ the cost is

$$C(z) = \sum_{(i,j)\in E} \frac{1 - z_i z_j}{2}. \tag{1}$$

The problem is NP-hard; the classical Goemans–Williamson algorithm guarantees a ratio of about 0.878 of the optimum (Goemans and Williamson, 1995).

1.2 QAOA

QAOA (Farhi et al., 2014) prepares

$$|\gamma, \beta\rangle = \prod_{l=1}^{p} e^{-i\beta_l B} e^{-i\gamma_l C} |+\rangle^{\otimes n}, \qquad B = \sum_j X_j, \tag{2}$$

and maximises $F_p(\gamma, \beta) = \langle \gamma, \beta | C | \gamma, \beta \rangle$ with a classical optimiser. Quality is measured by the approximation ratio $r = F_p / C_{\max}$. The landscape has many local optima, so initial angles matter.

1.3 Angle concentration and transfer

Brandao et al. (2018) showed that for fixed angles the QAOA objective concentrates over typical instances of a graph family, so good angles for one graph are good for similar graphs. Zhou

et al. (2020) found smooth patterns in optimal angles across depths and used them to build initialisations for larger p. Egger et al. (2021) warm-start QAOA from a classical relaxation. Learning a map from graph features to angles generalises these ideas with data.

2 Research questions

The project is open-ended: these questions are the starting point, not the finish line. Add new ones to the README and the journal as they come up.

RQ1 Which graph features (size, density, degree statistics, clustering) predict good angles?

RQ2 Do angles learned on 6–8-node graphs transfer to 10–12-node graphs?

RQ3 How much do predicted angles reduce optimiser steps compared with random and with fixed mean angles?

3 Method

3.1 Data

Random graphs generated with NetworkX (Hagberg et al., 2008): 3-regular graphs and Erdős–Rényi graphs with $p_{\text{edge}} \in \{0.3, 0.5\}$, 6 to 12 nodes. For each graph, the best angles for depth $p = 1, 2$ are found by optimisation from 20 random restarts; these are the regression targets. Angles are mapped to a canonical range to remove symmetries before training.

3.2 Models

Regressors from graph features to (γ, β): ridge regression, random forest and a small MLP. QAOA itself is simulated exactly with PennyLane; optimisation of the angles uses Adam or COBYLA with a fixed step budget.

3.3 Baselines

Random initial angles; fixed angles equal to the mean of the training targets (pure concentration); the INTERP heuristic of Zhou et al. (2020) for $p > 1$; Goemans–Williamson for the cut quality reference.

3.4 Metrics

- Approximation ratio at initialisation and after K optimiser steps.
- Steps needed to reach 95 % of the best-known ratio.
- Angle prediction error on held-out graphs.

4 Planned experiments

Each experiment becomes one `[[experiments]]` entry in the workspace, so its runs, parameters and metrics are stored and compared in the portal.

ID	Experiment	What varies	Main metrics
E1	Data set	graph family, $n = 6..10$, $p = 1$	target quality
E2	Regressors	ridge, forest, MLP	init ratio, steps saved
E3	Size transfer	train $n \leq 8$, test $n = 10, 12$	init ratio
E4	Deeper QAOA	$p = 2, 3$	steps saved vs INTERP
E5	Graph families	regular vs Erdős–Rényi	transfer across families

Table 1: Experiment plan. Results go into the journal and the results log (section 9).

5 Running it in the QML Lab

The workspace lives in `qml_projects/qaoa-maxcut-learned-init/`. Scripts import three helpers from `qml_lab/plots.py`:

- `params()` reads the parameters from the portal's run form (defaults when run by hand);
- `save()` stores figures with the run;
- `log_metrics()` records numbers that appear in the run table.

```
qml_projects/qaoa-maxcut-learned-init/
  project.toml          metadata + experiments
  README.md             goal, open questions, ideas
  experiments/          runnable scripts
  docs/main.tex         this document
```

First experiment: This experiment is *planned*: write the script, add the block below to `project.toml`, then run `python manage.py sync_projects`.

```
[[experiments]]
slug = "learned-init"
title = "Learned initial angles for depth-1 QAOA"
script = "experiments/e01_learned_init.py"
figure = "steps_saved.png"

[experiments.params]
n_nodes = 8
graph_type = "regular3"
depth_p = 1
n_graphs = 200
restarts = 20
seed = 0
```

```
# from the portal folder: pick up changes to project.toml
python manage.py sync_projects
cd qml_projects/qaoa-maxcut-learned-init
# run by hand with the default parameters
PYTHONPATH=../../qml_lab python experiments/e01_learned_init.py
```

6 Quality bar

- A classical baseline with a similar number of trainable parameters, tuned with the same care (Bowles et al., 2024).
- At least five random seeds per configuration, reported as mean $\pm$ standard deviation.
- How results change with the number of qubits.
- At least one run under a realistic noise model.
- Report the classical Goemans–Williamson ratio next to every QAOA result.

7 Open directions

- Graph neural networks as the angle predictor.
- Weighted MaxCut and other QUBO problems.
- Warm-starting from classical relaxations (Egger et al., 2021).
- Run predicted-angle QAOA on hardware (Project 16).

8 Risks and limitations

- Angle symmetries make targets ambiguous unless they are canonicalised.
- Small graphs are easy classically; the study is about optimisation cost, not about beating classical solvers.
- Generating targets needs many simulations; keep graphs small at first.

9 Results log

Summarise findings here as they become stable; day-to-day notes belong in the portal journal.

Date	Run	Finding	Next step

References

Bowles, J., Ahmed, S. and Schuld, M. (2024) 'Better than classical? The subtle art of benchmarking quantum machine learning models', Preprint, arXiv:2403.07059. Available at: https://arxiv.org/abs/2403.07059.

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Hagberg, A. A., Schult, D. A. and Swart, P. J. (2008) 'Exploring network structure, dynamics, and function using NetworkX', in *Proceedings of the 7th Python in Science Conference (SciPy 2008)*, pp. 11–15.

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